



Quantum Machine Learning for Small Datasets

*How Far Does the Quantum Advantage Hold Up
in Accuracy, Execution Time, and Practical Usability?*

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Introduction & Motivation

Why Quantum Computing? Why Now?

In 2019, Google announced quantum supremacy with the Sycamore processor -- performing in minutes a task estimated to take 10,000 years on a classical supercomputer. This sparked commercial investment from IBM, Amazon, Microsoft, and D-Wave.

Today, cloud-accessible quantum computers are a reality. IBM's platform offers machines with up to 156 qubits free for academic research.

Yet the key question remains unanswered in practical terms:

"Can quantum ML models outperform classical ML in real-world, small-dataset scenarios?"

This dissertation benchmarks 4 classical vs 7 quantum/hybrid ML models on the Iris dataset using IBM's real quantum hardware (QPU).

2019

Google Quantum
Supremacy

156

IBM Qubits
Available

11

Classical + Quantum
Models Tested

486

Quantum Circuits
Executed

RQ 1 -- Accuracy

Does quantum machine learning achieve competitive predictive accuracy compared to classical ML on the Iris dataset?

RQ 2 -- Execution Time

What is the computational cost (training time and inference time) of quantum models relative to classical baselines?

RQ 3 -- Practical Usability

How usable are current quantum ML frameworks for practitioners in terms of reliability, tooling, and real QPU execution?

Hypotheses

H0 (Null): No significant difference in accuracy between classical and quantum models.

H1 (Alternative): Quantum models achieve statistically competitive accuracy, with trade-offs in computational cost and usability.

Scope: Iris dataset (150 samples, 3 classes) | IBM Quantum (Heron r2 QPU, 156 qubits) | Qiskit ML | SciKit-Learn | 2-qubit classification

Quantum Computing Fundamentals

The building blocks that make quantum ML possible

Qubits

Unlike classical bits (0 or 1), qubits exist in superposition -- both states simultaneously -- until measured. This allows exponential parallelism in certain computations.

Quantum Entanglement

Two qubits can be entangled so that measuring one instantly determines the other, regardless of distance. Einstein called it "spooky action at a distance."

Superposition & Coherence

A qubit in superposition holds multiple states until observed. Quantum coherence -- maintaining this state without external interference -- is a major engineering challenge in the NISQ era.

Josephson Junctions & Gates

IBM's superconducting qubits use Josephson junctions cooled near absolute zero. Quantum gates manipulate qubit states through controlled microwave pulses, analogous to classical logic gates.

Key milestone: Google (2019) demonstrated quantum supremacy. Google Willow (2024) showed error suppression below the surface-code threshold for the first time.

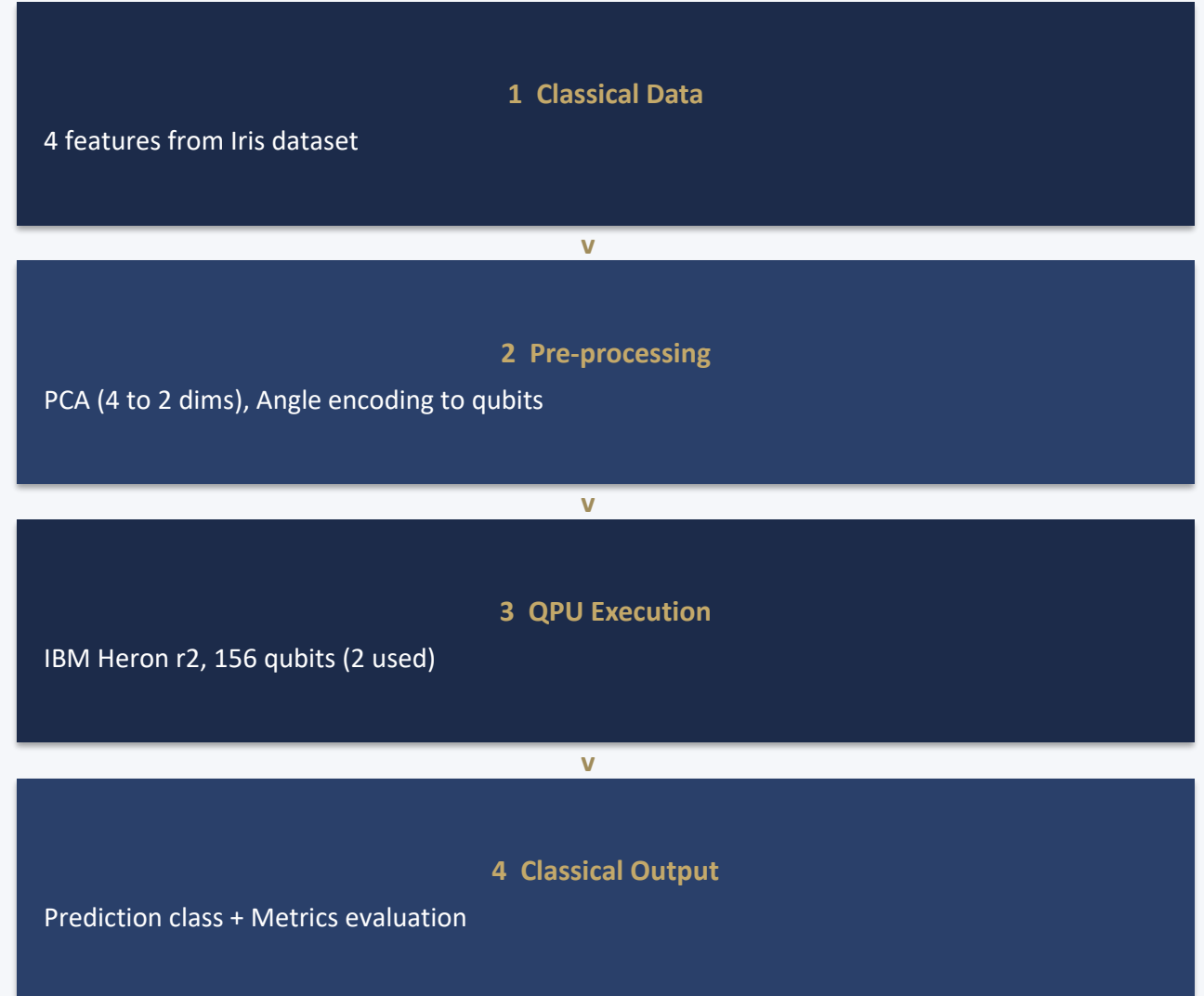
Quantum Machine Learning (QML)

Combining quantum computation with machine learning paradigms

QML, formalised by Biamonte et al., distinguishes between approaches that accelerate linear algebra (data-centric) and trainable parametrised circuits (model-centric).

Variational Quantum Circuits (VQCs) -- the NISQ-era workhorse -- use classical optimisers to tune quantum circuit parameters, analogous to gradient descent in neural networks.

Data is encoded via angle encoding: classical features are mapped to qubit rotation angles using MinMaxScaler in the range $[0, \pi]$ (Bloch sphere).



Experimental Setup

IBM Quantum Platform | Qiskit ML | Real QPU Hardware

156

IBM Heron r2 QPU
qubits available

2

Qubits used
in classification

486

Quantum circuits
executed

33/33

Jobs completed
(zero failures)

Hardware & Software Stack

IBM Quantum Platform (Heron r2 QPU, superconducting)
Qiskit 1.x | Qiskit Machine Learning
SciKit-Learn | Python 3.11 | AerSimulator (local testing)
Config via .env (IBM_BACKEND_NAME, TRAIN_SPLIT, shots)

Execution Conditions

All 11 models ran in the same execution context
Queue wait of approx. 843 s dominated wall-clock time
Shots per circuit: ~1024 (probabilistic machine)
Source code: github.com/wevertonsc/TUS_DISSERTATION_2026

Dataset & Pre-Processing

Iris -- the benchmark that launched a thousand classifiers

The Iris dataset (Ronald Fisher, 1936) is one of the most widely used benchmarks in classification literature. It was chosen for its broad comparability across classical and quantum studies.



Pre-Processing Pipeline



Models Evaluated

4 Classical Baselines vs 7 Quantum / Hybrid Models

Classical (SciKit-Learn)

SVM

Support Vector Machine -- radial basis function kernel, C=1

k-NN

k-Nearest Neighbours -- k=3, Euclidean distance

LogReg

Logistic Regression -- L2 penalty, max_iter=200

MLP

Multi-Layer Perceptron -- (64,32), ReLU, Adam

Quantum / Hybrid (Qiskit ML)

QSVM

Quantum kernel SVM -- quantum fidelity kernel (ComputeUncompute)

QkNN

Quantum k-NN -- fidelity-based similarity (proxy: Gaussian kernel)

QCL

Quantum Circuit Learning -- ZZFeatureMap + RealAmplitudes + COBYLA

VQC

Variational Quantum Classifier -- ZZFeatureMap + COBYLA

QNN

Quantum Neural Network -- EstimatorQNN + COBYLA

QFC

Quantum Feature Classifier -- fixed non-variational circuit

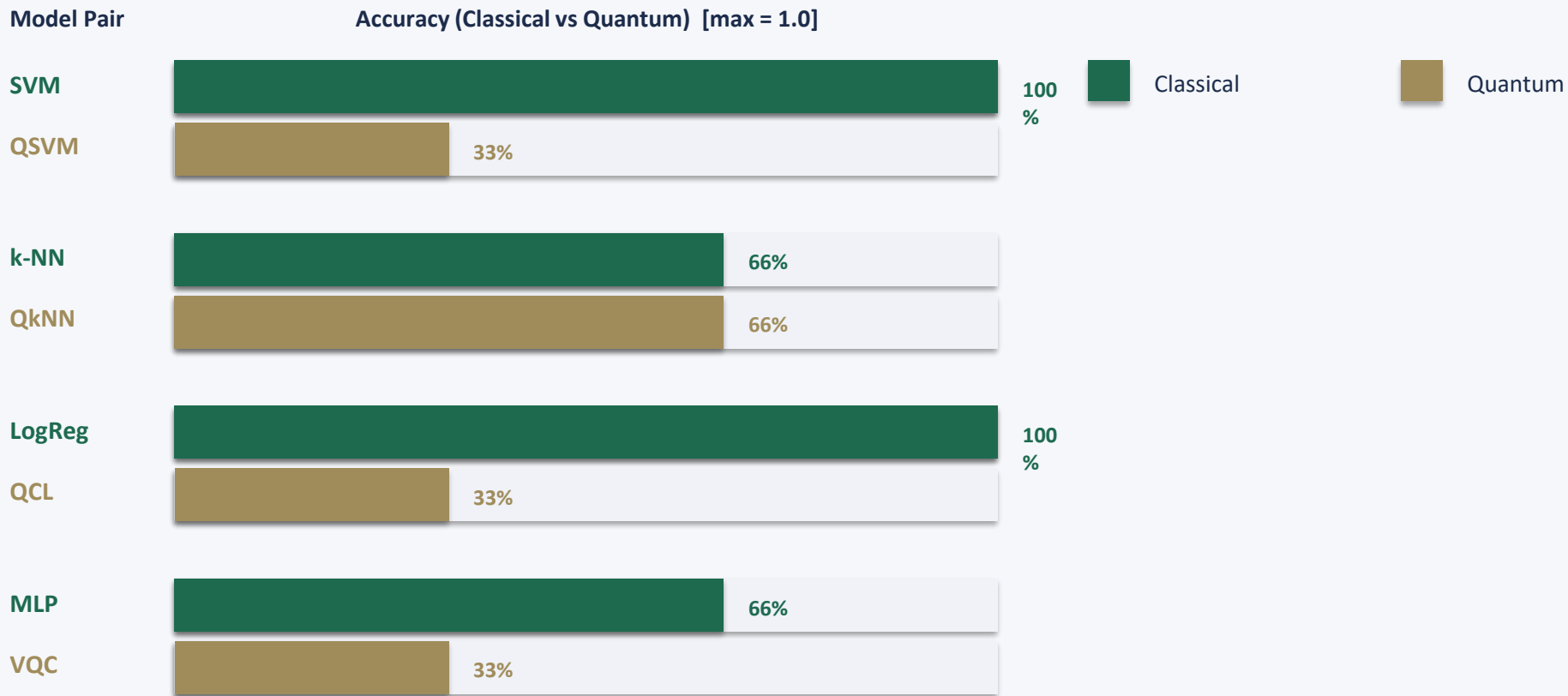
QFE-LR

Quantum Feature Extraction + Logistic Regression head

Results -- Predictive Performance

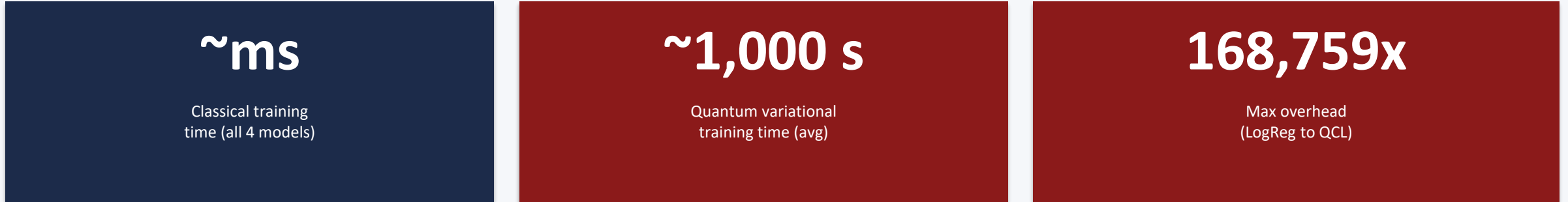
Classical models equal or outperform quantum in every paired comparison

Classical accuracy advantage confirmed | H0 rejected | Logistic Regression: 100% accuracy in milliseconds | Best quantum model (QkNN 66.7%) ran on classical backend -- not on QPU

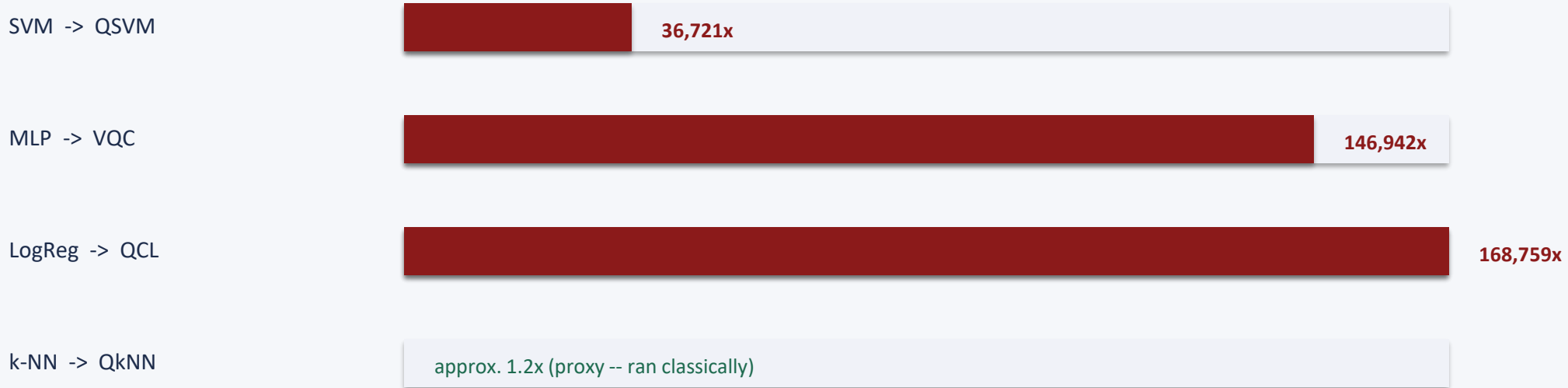


Results -- Computational Cost

Quantum training time: orders of magnitude above classical baselines



Training Time Ratio (Quantum / Classical)



Note: wall time includes approx. 843 s queue latency on IBM QPU. QkNN best-quantum result ran on a classical Gaussian kernel (proxy) -- not actual QPU.

Discussion & Limitations

Context matters: NISQ era constraints shaped every result

Why Did Quantum Underperform?

Small test set

Only 6 test samples (2 per class) -- each miss shifts accuracy by 0.167

Dimensional reduction

PCA to 2 components discards information and merges Versicolor / Virginica classes

8-iteration ceiling

Hardware guards limited COBYLA to 8 iterations -- variational circuits never converged

Queue latency

~843 s wait dominated wall time; pure QPU compute time was far shorter

QkNN proxy

Best quantum result (66.7%) ran on a classical kernel -- not on QPU hardware

Broader Context & Positive Signals

Hardware reliability

33/33 jobs completed with zero failures across 486 circuits -- QPU stability is real.

Variational overhead

Variational models averaged 7.7x slower than non-variational quantum models -- a known NISQ bottleneck.

Commercial signals

HSBC (2025) reported 34% improvement in bond trading using quantum-assisted algorithms on IBM.

Error correction progress

Google Willow (2024) demonstrated exponential error suppression below the surface-code threshold.

Key Findings

- H0 rejected -- classical models outperformed quantum in every pair
- Logistic Regression: 100% accuracy in milliseconds vs QCL: 33% in >1,000 s
- Overhead: 36,721x (SVM->QSVM) to 168,759x (LogReg->QCL)
- Variational circuits failed to converge under 8-iteration ceiling
- QPU reliability: 33/33 jobs, zero failures -- hardware is stable

Future Work

Larger datasets

Test on datasets with >1,000 samples where quantum feature maps may generalise better

More qubits

Explore 4-8 qubit circuits as QPU error rates decrease over time

Error mitigation

Apply ZNE / Pauli twirling to reduce NISQ noise impact on accuracy

Hybrid pipelines

Quantum feature extraction combined with classical deep-learning heads

Reproducibility

All source code available:
github.com/wevertonsc/TUS_DISSERTATION_2026